

## Working Principal, Construction Of Single phase and Three Phase Induction Motor

### CONSTRUCTION OF SINGLE PHASE INDUCTION MOTOR

For general lighting purpose in shops, offices, houses, schools etc. Single phase a.c. supply is commonly used. Hence instead of d.c. motors, the motors which work on single phase a.c. supply are very popularly in use. These a.c. motors are called single phase induction motors. The numerous domestic applications use single phase motors. The power rating of such motors is very small. Some of them are even fractional horse power motors, which are used in applications like small toys, small fans, hair dryers etc. This chapter explains the construction, working principle and applications of various types of single phase induction motors.

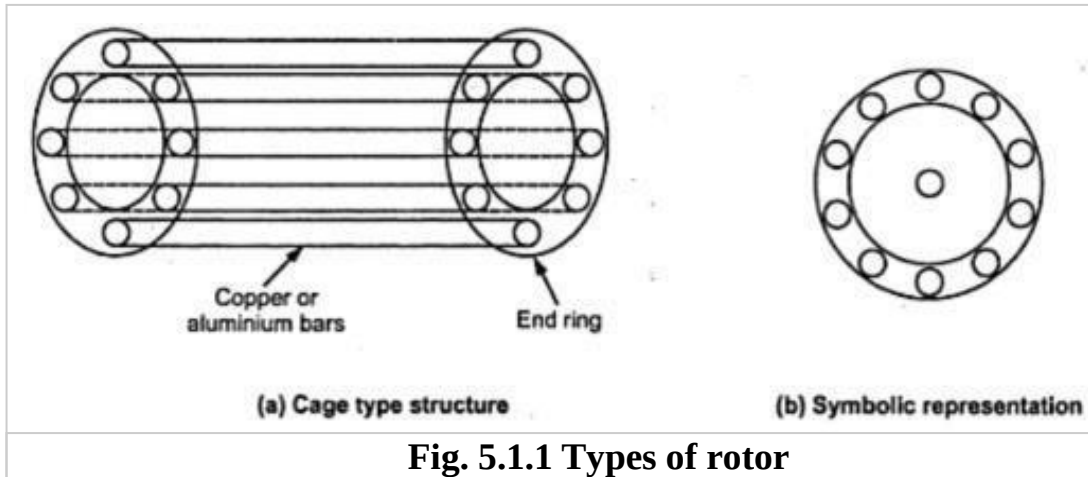
Similar to a d.c. motor, single phase induction motor has basically two main parts, one rotating and other stationary. The stationary part in single phase induction motors is called stator while the rotating part is called rotor.

The stator has laminated construction, made up of stampings. The stampings are slotted on its periphery to carry the winding called stator winding or main winding. This is excited by a single phase a.c. supply. The laminated construction keeps iron losses to minimum. The stampings are made up of material like silicon steel which minimizes the hysteresis loss. The stator winding is wound for certain definite number of poles means when excited by single phase a.c. supply, stator produces the magnetic field which creates the effect of certain definite number of poles. The number of poles for which stator winding is wound, decides the synchronous speed of the motor. The synchronous speed is denoted as  $N_s$  and it has a fixed relation with supply frequency  $f$  and number of poles  $P$ . The relation is given by,

$$N_s = \frac{120 f}{P} \text{ r.p.m}$$

The induction motor never rotates with the synchronous speed but rotates at a speed which is slightly less than the synchronous speed.

The rotor construction is of squirrel cage type. In this type, rotor consists of uninsulated copper or aluminium bars, placed in the slots. The bars are permanently shorted at both the ends with the help of conducting rings called end rings. The entire structure looks like cage hence called squirrel cage rotor. The construction and symbol is shown in the Fig.5.1



As the bars are permanently shorted to each other, the resistance of the entire rotor is very very small. The air gap between stator and rotor is kept uniform and as small as possible. The main feature of this rotor is that it automatically adjusts itself for same number of poles as that of the stator winding.

The schematic representation of two pole single phase induction motor is shown in the Fig.5.2.

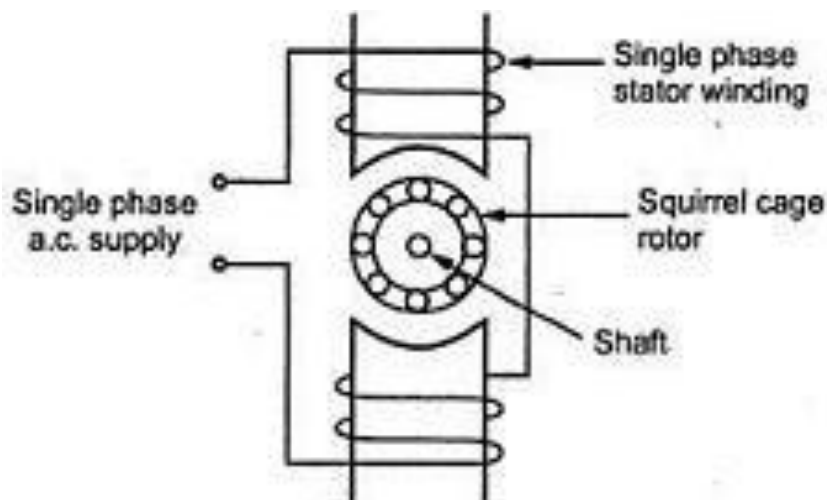


Fig 5.2 construction

### **Working Principle of 1-phase Induction Motor**

For the motoring action, there must exist two fluxes which interact with each other to produce the torque. In d.c. motors, field winding produces the main flux while d.c. supply given to armature is responsible to produce armature flux. The main flux and armature flux interact to produce the torque.

In the single phase induction motor, single phase a.c. supply is given to the stator winding. The stator winding carries an alternating current which produces the flux which is also alternating in nature. This flux is called main flux. This flux links with the rotor conductors and due to transformer action e.m.f. gets induced in the rotor. The induced e.m.f. drives current through the rotor as rotor circuit is closed circuit. This rotor current produces another flux called rotor flux required for the motoring action. Thus second flux is produced according to induction principle due to induced e.m.f. hence the motor is called induction motor. As against this in d.c. motor a separate supply is required to armature to produce armature flux. This is an important difference between d.c. motor and an induction motor.

Another important difference between the two is that the d.c. motors are self starting while single phase induction motors are not self starting.

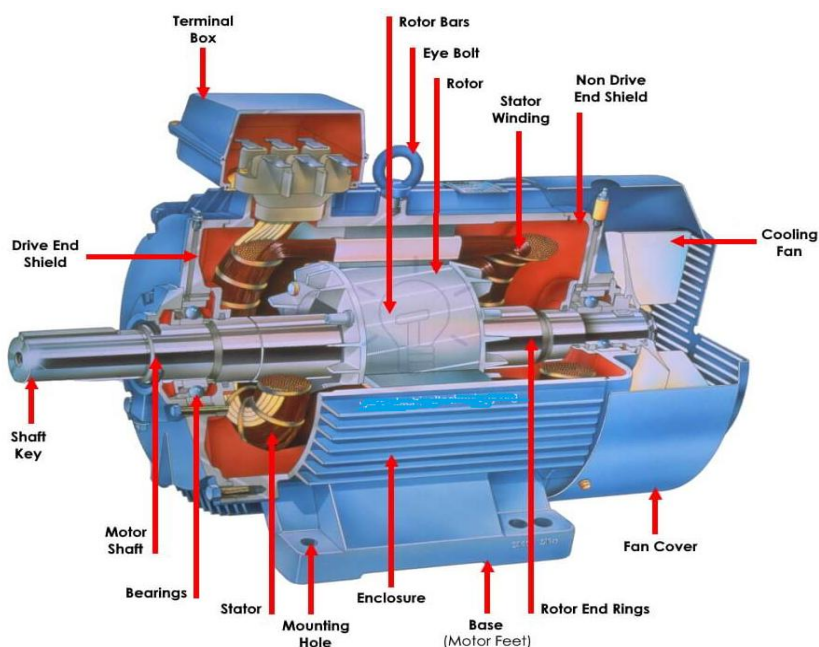
Let us see why single phase induction motors are not self starting with the help of a theory called double revolving field theory.

### Three Phase Induction Motor

The three phase induction motor is one of the A.C. motors, which is widely used for various purposes in industry. These motors never run at a Synchronous speed but a little less than the synchronous speed. The speed of these motors depends upon the supply frequency.

Therefore, these motors are not generally used for speed control. However, we prefer D.C. motors where large variations of speed are required. These motors are preferred in industry because they have low price, simple & rugged construction, can be manufactured with characteristics to suit the industrial requirement.

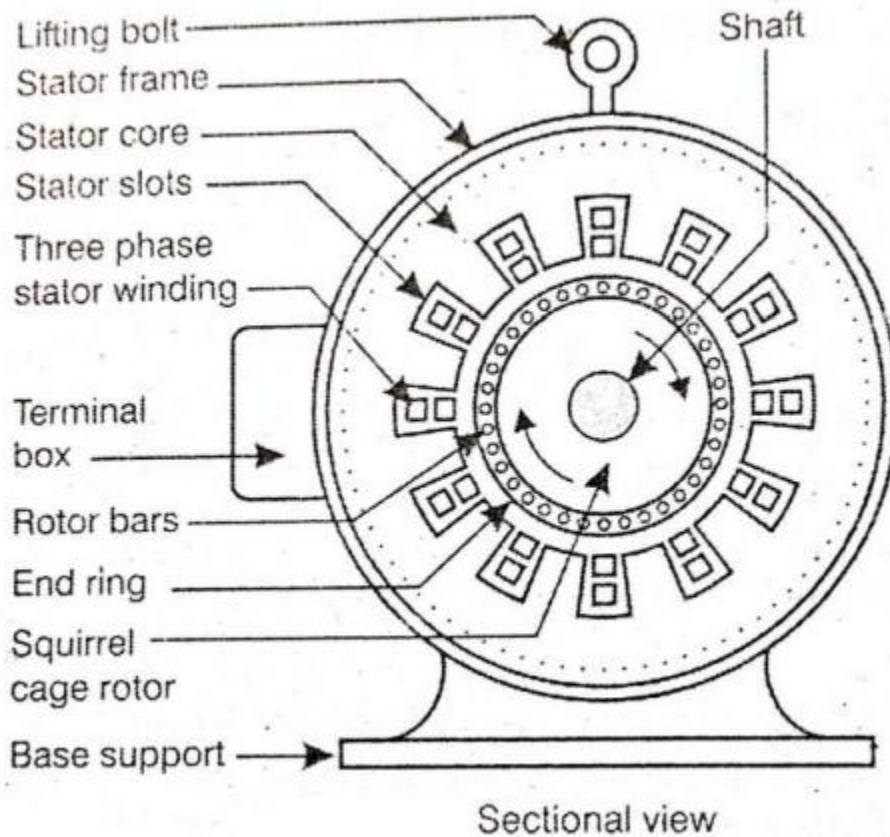
These motors differ from other types of motor, in that there is no electrical connection between the rotor & supply. The required voltage & current are induced by induction from the **stator winding** that is why, the name given is induction motor.



### Construction of Three Phase Induction Motor

It can be better understood if we see the construction of three phase induction motor which has two major parts :

1. Stationary part, known as Stator
2. Rotating part, known as Rotor.



## 1. Stator

It is the stationary part of the motor. It has three main parts :

### **Frame or Yoke**

It is the outer part of the three phase induction motor. Its main function of the frame is to support the stator core & stator winding. It acts as a covering, and it provides protection & mechanical strength to all the inner parts of the three phase induction motor.

### **Stator core**

The main function of stator core is to carry the alternating flux. In order to reduce the eddy current loss, the stator core is laminated. The core is made up of thin silicon steel laminations. These are insulated from each other by varnish, the slots are cut on inner periphery of core stampings. The stator windings are placed in these slots.

### **Stator windings**

Stator winding is made up of super enamelled copper wire. Three phase windings are placed in the stator core slots & six terminals are brought out. They may be star connected or may be delta connected. The windings are connected in star at starting.

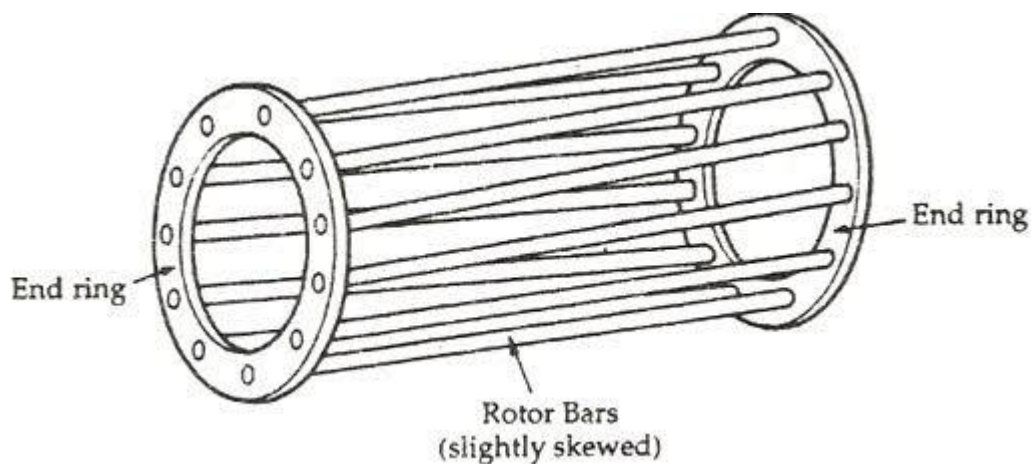
## 2. Rotor

It is a rotating part of the motor. It is mounted on the shaft. It consists of hollow laminated core having slots on its outer periphery. The windings placed in these slots (rotor winding) may be one of the following two types :

1. Squirrel cage rotor
2. Slip ring rotor or wound rotor or phase wound rotor.

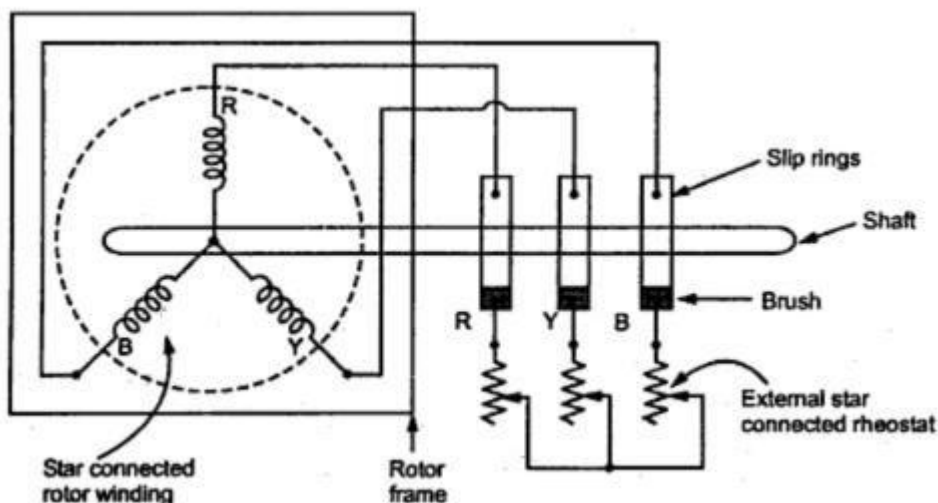
### 1. Squirrel cage rotor

The rotor consists of a cylindrical laminated core with parallel slots for carrying the rotor conductors. The **squirrel cage** rotor consists of a aluminium, brass or copper bars. These aluminium, brass or copper bars are called rotor conductors & are placed in the slots on the periphery of the rotor. The rotor conductors are permanently shorted by the copper, or aluminum rings called the **end rings**. To provide mechanical strength, these rotor conductors are braced to the end ring & hence form a complete closed circuit resembling like a cage & hence got its name as squirrel cage induction motor.



### 2. Slip ring rotor or wound rotor or phase wound rotor

The wound rotor consists a slotted armature. Insulated conductors are put in the slots & connected to form a three phase double layer distributed winding similar to the stator winding. The rotor windings are connected in star.



The open end of the start circuit are brought outside the rotor and connected to the insulated slip rings. The slip rings are mounted on the shaft with brushes resting on them. The brushes are connected to three phase variable resistors connected in star. The purpose of slip rings & brushes is to provide a means for connecting external resistors in the circuit.

### **Principle of Operation of 3-Phase Induction Motor**

The three phase induction motor works on the principle of electromagnetic induction. When a three-phase supply is given to three-phase winding of the motor, a magnetic field is produced which rotates at synchronous speed. The rotating **flux** passes through the air gap and cuts the rotor conductors which are at rest. The rotor winding is short circuited, therefore, the current will flow in the rotor winding due to induced emf & a magnetic field is setup.

Now these two **magnetic field** interact, a torque is produced. According to **Lenz's law** under the influence of this torque, the rotor starts rotating in the same direction as the rotating magnetic field.

The speed of the rotor will be always less than the speed of the field. The emf in the rotor is induced by the law of **electromagnetic induction**, therefore, this motor is called induction motor.